AI ENGINEERING / OPERATIONS / FINANCE

DELIVERY MATCHED INVOICING

09

PROPOSED RESEARCH / NOT YET VERIFIED

THE ENGINEERING QUESTION

How reliably can invoices be matched to what was actually received so rejected goods are credited without wrongly short paying suppliers?

PROBLEM DEFINITION AND RESEARCH CONTRIBUTION

A proposed matching workflow that compares supplier invoices with goods received notes and rejection records then drafts credit note requests for a person to approve.

- Match at line level using received quantities and rejection reasons rather than whole invoice totals.
- Keep the supplier relationship in view by drafting factual requests that cite the record behind each line.
- Use synthetic suppliers and deliveries only and never imply recovered value or live results.

ARCHITECTURE AND IMPLEMENTATION SPECIFICATION

1 STAGE **Invoice intake** → 2 STAGE **Receipt matching** → 3 STAGE **Credit draft** → 4 STAGE **Buyer approval**

- Typed records for purchase orders, goods received notes, temperature logs and rejection reasons linked by line.
- Matching engine handling partial deliveries and unit conversions with an explicit confidence and reason for every match.
- Credit note request drafts with line evidence plus an approval log recording who approved edited or declined each draft.

ENGINEERING DECISIONS TO MAKE EXPLICIT

The design review must record data contracts, state transitions, failure handling and why the selected baseline is defensible. Every subsystem should have an interface specification and a reproducible test route.

DESIGN BOUNDARIES AND ASSUMPTIONS

⚠ Synthetic deliveries cannot show how real sites record rejections or how suppliers answer credits.

⚠ A missing rejection record is a question for a person and never proof of acceptance.

ⓘ Scope is proposed unless explicitly identified as documented existing work. All measured claims require reproducible evidence.

DELIVERY MATCHED INVOICING / PROTOCOL / ACCEPTANCE / REVIEW

EXPERIMENTS, VERIFICATION & PUBLICATION

09

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EXPERIMENTAL METHODOLOGY

01 Generate synthetic orders and receipts with labelled temperature and quality rejections and invoices that ignore them.

02 Compare manual checking, rules only matching and a hybrid that uses a model only to read free text rejection notes.

03 Measure credit recall and precision and short payment risk and approver edit rate on held out synthetic batches.

MEASUREMENT AND ACCEPTANCE EVIDENCE

- Credit value found vs injected truth on synthetic deliveries
- Match rate across partial deliveries and substituted lines
- Wrongly drafted credits that a reviewer would reject

REQUIRED REPOSITORY ARTEFACTS

- Synthetic orders receipts and invoice dataset
- Matching engine with line level test suite
- Credit request drafts and evaluation report

TECHNICAL REVIEW QUESTIONS

Which assumptions could invalidate the result? How do the baselines behave under the same conditions? What is the strongest negative case? Which variables, seeds, workloads, software versions and machine properties must be retained for independent reproduction?

VERIFICATION MAPPING

For each acceptance claim, record the test identifier, input configuration, observed result, pass/fail definition and an independent rerun procedure. Failed or inconclusive cases remain in the report rather than being silently excluded.

PUBLICATION DISCIPLINE

Release original source, configuration files, test harnesses, raw measurements and scripts where rights permit. Distinguish simulation outcomes from physical validation and completed achievements from planned evaluation. Results must not be inferred from targets.